curl -fsSL https://get.docker.com -o get-docker.sh
sh get-docker.sh

```
root@debianMSI:~# docker ps .a
docker: 'docker ps' accepts no arguments

Usage:  docker ps [OPTIONS]

Run 'docker ps --help' for more information
root@debianMSI:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@debianMSI:~# dmesg | grep -i \"driver\|firmware\"
root@debianMSI:~# lspci | grep VGA
2b:00.0 VGA compatible controller: NVIDIA Corporation GA104 [GeForce RTX 3060 Ti GDDR6X] (rev a1)
root@debianMSI:~#
```

Como se va usar una nvidea 3060TI

1) Driver NVIDIA + toolkit para Docker

(para tu 3060 Ti; reinicia una vez en medio)

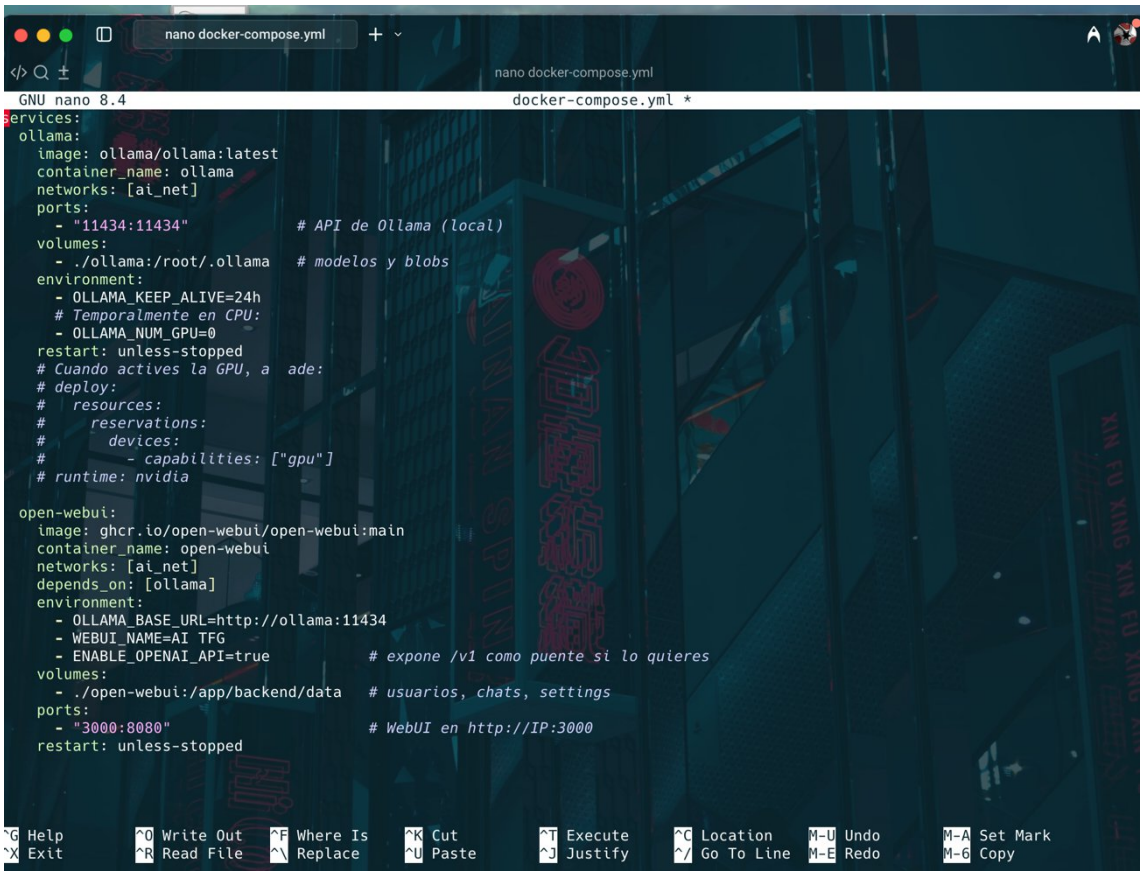
```
# habilita non-free-firmware
sudo sed -i 's/main/main contrib non-free non-free-firmware/g'
/etc/apt/sources.list
sudo apt update
sudo apt install -y linux-headers-$(uname -r) nvidia-driver
sudo reboot
```

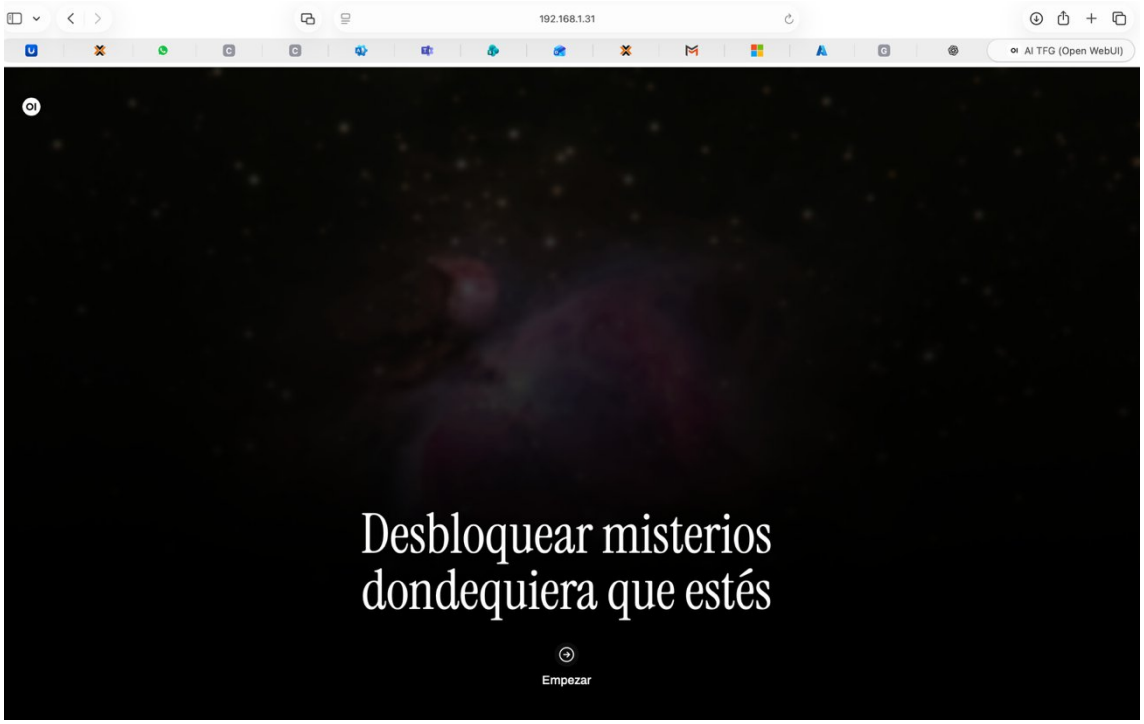
```
cmateos@debianMSI ~  (0.016s)
mkdir -p ~/tfg/{nginx/conf.d,api,webui,qdrant_data,uptime-kuma,zbx_db}
cd ~/tfg

cmateos@debianMSI ~/tfg  (0.014s)
pwd
/home/cmateos/tfg

cmateos@debianMSI   ~/tfg

gpt-4o
```


nano docker-compose.yml
nano docker-compose.yml
GNU nano 8.4
docker-compose.yml *
services:
ollama:
image: ollama/ollama:latest
container_name: ollama
networks: [ai_net]
ports:
- "11434:11434" # API de Ollama (local)
volumes:
- ./ollama:/root/.ollama # modelos y blobs
environment:
- OLLAMA_KEEP_ALIVE=24h
Temporalmente en CPU:
- OLLAMA_NUM_GPU=0
restart: unless-stopped
Cuando actives la GPU, a ade:
deploy:
resources:
reservations:
devices:
- capabilities: ["gpu"]
runtime: nvidia
open-webui:
image: ghcr.io/open-webui/open-webui:main
container_name: open-webui
networks: [ai_net]
depends_on: [ollama]
environment:
- OLLAMA_BASE_URL=http://ollama:11434
- WEBUI_NAME=AI TFG
- ENABLE_OPENAI_API=true # expone /v1 como puente si lo quieres
volumes:
- ./open-webui:/app/backend/data # usuarios, chats, settings
ports:
- "3000:8080" # WebUI en http://IP:3000
restart: unless-stopped
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy


192.168.1.31
AI TFG (Open WebUI)
Desbloquear misterios
dondequiera que estés
Empezar

Empezar con AI TFG (Open WebUI)

ⓘ AI TFG (Open WebUI) no se realiza ninguna conexión externa y tus datos permanecen seguros alojados localmente en tu servidor.

Nombre

Carlos

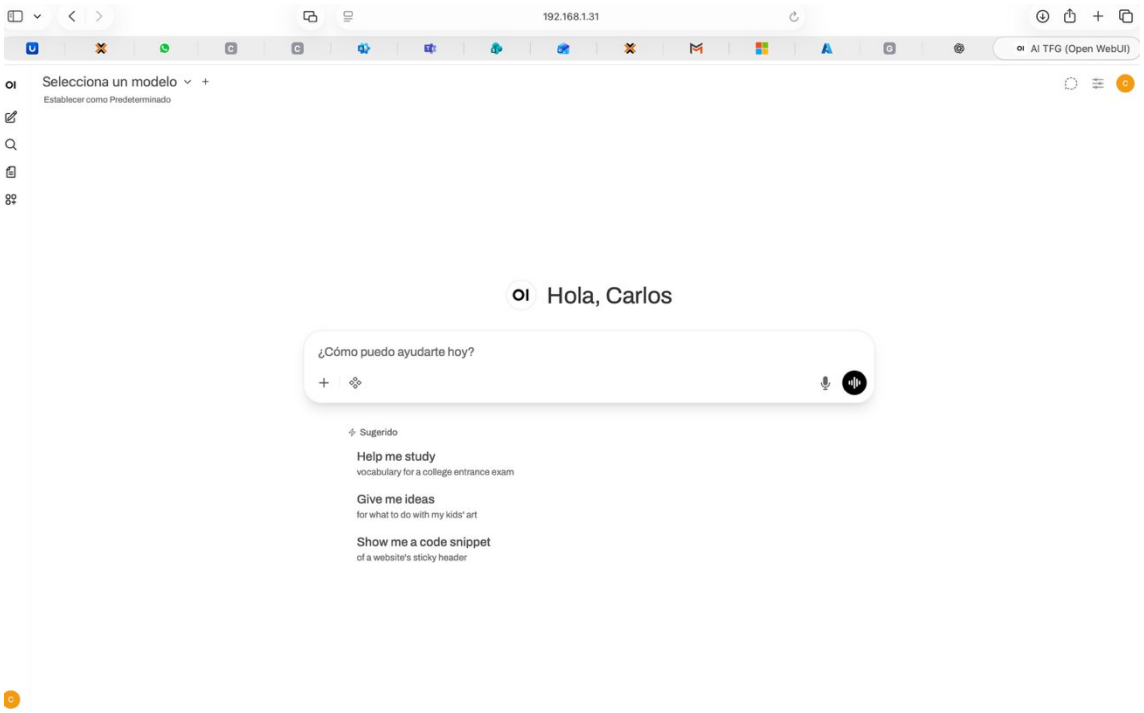
Email

cmateos@cmateos.com

Contraseña

••••••••

Crear Cuenta Administrativa



Selecciona un modelo

llama3.1

No se encontraron resultados

Downloading "llama3.1" (31%)
sha256:667b0c1932bc6ffc593ed1d03f895bf2dc8dc6df21db3042284a6f...

debianMSI:~/tfg

```
cmateos@debianMSI ~/tfg  (1.109s)
sudo apt install htop

Get:1 http://deb.debian.org/debian trixie/main amd64 htop amd64 3.4.1-5 [171 kB]
Fetched 171 kB in 0s (5863 kB/s)
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = "UTF-8",
        LC_NUMERIC = (unset),
        LC_COLLATE = (unset),
        LC_TIME = (unset),
        LC_MESSAGES = (unset),
        LC_MONETARY = (unset),
        LC_ADDRESS = (unset),
        LC_IDENTIFICATION = (unset),
        LC_MEASUREMENT = (unset),
        LC_PAPER = (unset),
        LC_TELEPHONE = (unset),
        LC_NAME = (unset),
        LANG = "es_ES.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("es_ES.UTF-8").
locale: Cannot set LC_CTYPE to default locale: No such file or directory
locale: Cannot set LC_ALL to default locale: No existe el fichero o el directorio
Selecting previously unselected package htop.
(Reading database ... 68692 files and directories currently installed.)
Preparing to unpack .../htop_3.4.1-5_amd64.deb ...
Unpacking htop (3.4.1-5) ...
Setting up htop (3.4.1-5) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for man-db (2.13.1-1) ...
```

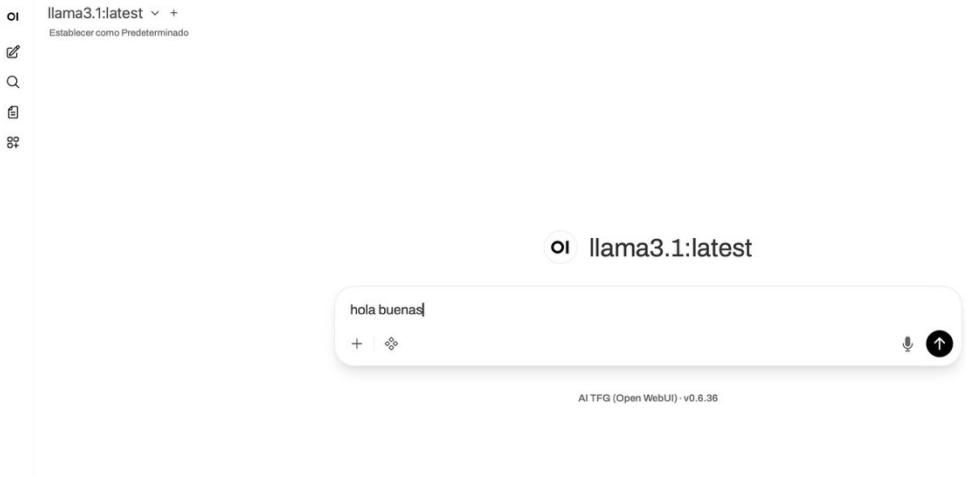
cmateos@debianMSI ~/tfg

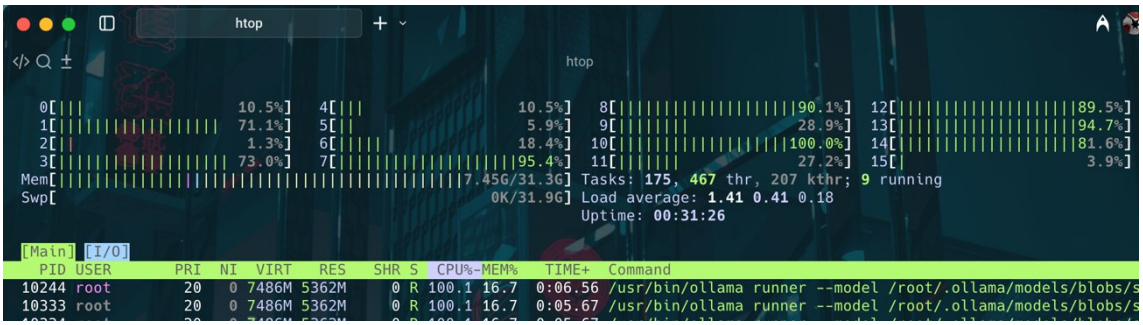
gpt-4o

htop

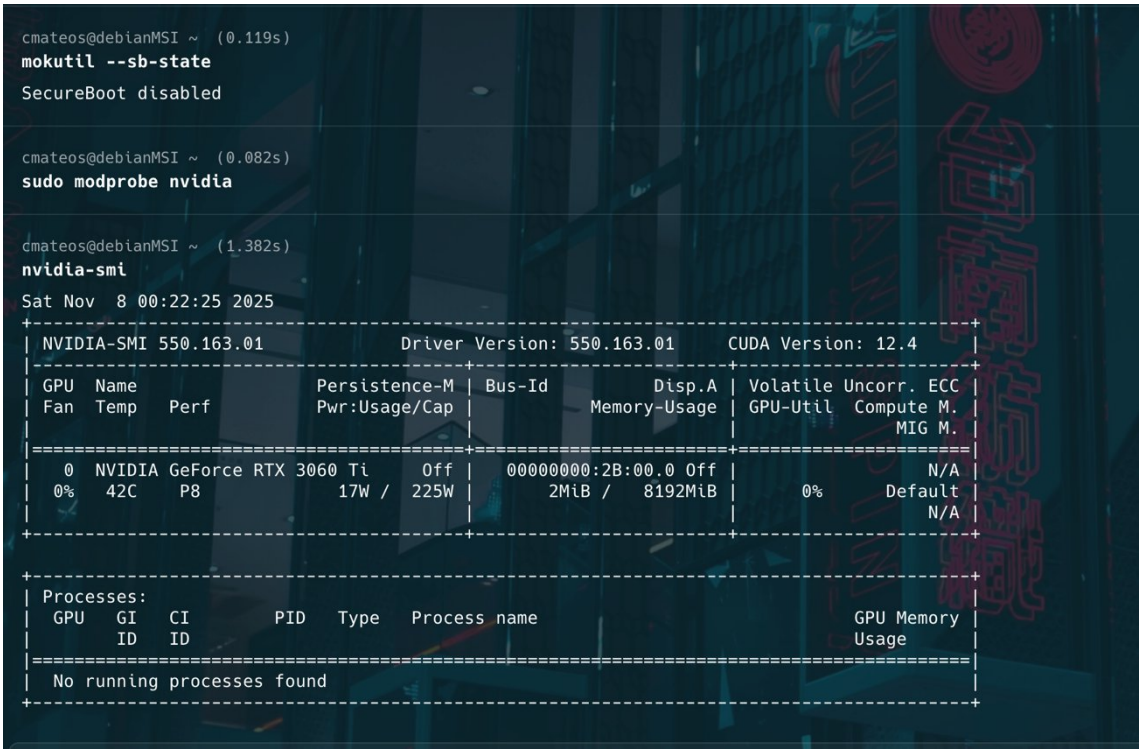
```
  0[                    0.0%]   4[                    0.0%]   8[                    0.0%]  12[                    0.0%]
  1[|                   0.7%]   5[                    0.0%]   9[||                  1.3%]  13[                    0.0%]
  2[|                   0.7%]   6[                    0.0%]  10[                    0.0%]  14[                    0.0%]
  3[|                   0.7%]   7[                    0.0%]  11[                    0.0%]  15[|                   0.7%]
Mem[||||||||||||||||||||||||||||||||||||||||||||  2.15G/31.3G] Tasks: 174, 449 thr, 205 kthr; 2 running
Swp[                                              0K/31.9G] Load average: 0.09 0.13 0.09
                                                            Uptime: 00:30:48

[Main] [I/O]
  PID USER       PRI  NI  VIRT   RES   SHR S  CPU%-MEM%   TIME+  Command
 4899 root        20   0  551M  333M     0 S   0.7  1.0  0:00.05 ./qdrant
 4949 root        20   0  551M  333M     0 S   0.7  1.0  0:00.04 ./qdrant
```





A continuación probamos GPU



Herramientas necesarias (te falta gnupg)

sudo apt update

```
sudo apt install -y curl ca-certificates gnupg lsb-release

# Clave GPG del repo
sudo install -m 0755 -d /usr/share/keyrings
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey \
 | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-container-toolkit.gpg

# Lista APT (canal estable compatible con Debian)
curl -fsSL https://nvidia.github.io/libnvidia-container/stable/deb/nvidia-container-toolkit.list \
 | sed 's#deb https://#deb [signed-by=/usr/share/keyrings/nvidia-container-toolkit.gpg] https://#' \
 | sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list

# (Opcional) añade también el canal experimental por si faltan dependencias
curl -fsSL https://nvidia.github.io/libnvidia-container/experimental/deb/nvidia-container-toolkit.list \
 | sed 's#deb https://#deb [signed-by=/usr/share/keyrings/nvidia-container-toolkit.gpg] https://#' \
 | sudo tee -a /etc/apt/sources.list.d/nvidia-container-toolkit.list

sudo apt update
sudo apt install -y nvidia-container-toolkit
```

Ahora mismo hemos conseguido instalar los drivers de nvidea en el equipo, y la IA ya está tirando de ella. Tenemos respuestas rápidas y bien formuladas.

fase 2 del TFG: integración, seguridad y presentación.

Intento de conectarse a la API y tener nuestra propia pagina

```
root@debianMSI:/home/cmateos/tfg#
root@debianMSI:/home/cmateos/tfg# nano api/main.py
api/                docker-compose.yml  ollama/             qdrant_data/        webui/
certs/              nginx/              open-webui/         uptime-kuma/        zbx_db/
root@debianMSI:/home/cmateos/tfg# nano api/main.py
root@debianMSI:/home/cmateos/tfg# curl -X POST http://localhost:8000/chat \
  -H "Content-Type: application/json" \
  -d '{"message": "Hola, ¿quién eres?"}'
curl: (7) Failed to connect to localhost port 8000 after 0 ms: Could not connect to server
root@debianMSI:/home/cmateos/tfg# cat api/main.py
from fastapi import FastAPI, Request
import requests

app = FastAPI()

OLLAMA_URL = "http://ollama:11434/api/chat"

@app.get("/status")
def status():
    return {"status": "ok", "service": "fastapi"}

@app.post("/chat")
def chat(request: dict):
    message = request.get("message", "")
    payload = {
        "model": "qwen2.5:7b",
        "messages": [{"role": "user", "content": message}]
    }
    r = requests.post(OLLAMA_URL, json=payload)
    return r.json()
root@debianMSI:/home/cmateos/tfg#
```

El API no funciona y debemos ver que está pasando

```
sudo su -
Your login has expired. Please sign in again to restore access to cloud-based features.  Sign in
sudo su -
  GNU nano 8.4                    /home/cmateos/tfg/api/main.py *
# /app/main.py  (montado desde ./api/main.py)
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
import requests
import os

app = FastAPI(title="TFG ASIR API")

OLLAMA_BASE = os.getenv("OLLAMA_BASE", "http://ollama:11434")

class ChatIn(BaseModel):
    message: str
    model: str | None = "qwen2.5:7b"

@app.get("/status")
def status():
    # prueba conexión a Ollama también
    try:
        r = requests.get(f"{OLLAMA_BASE}/api/tags", timeout=5)
        ok = r.status_code == 200
    except Exception:
        ok = False
    return {"status": "ok", "ollama_ok": ok}

@app.post("/chat")
def chat(body: ChatIn):
    payload = {
        "model": body.model,
        "messages": [{"role": "user", "content": body.message}],
    }
    try:
        r = requests.post(f"{OLLAMA_BASE}/api/chat", json=payload, timeout=60)
        r.raise_for_status()
        return r.json()
    except requests.exceptions.RequestException as e:
        # Devuelve 502 si Ollama no responde o hay error de red/HTTP

^G Ayuda      ^O Guardar    ^F Buscar     ^K Cortar     ^T Ejecutar   ^C Ubicación  M-U Deshacer  M-A Poner marca
^X Salir      ^R Leer fich. ^\ Reemplazar ^U Pegar      ^J Justificar ^/ Ir a línea M-E Rehacer  M-6 Copiar
```

root@debianMSI:/home/cmateos/tfg# docker compose restart api

o fuerza reinstalar deps si alguna falta

```
docker compose exec api pip install --no-cache-dir fastapi uvicorn requests
pydantic

WARN[0000] /home/cmateos/tfg/docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion

[+] Restarting 1/1

 ✔ Container fastapi  Started
0.5s

WARN[0000] /home/cmateos/tfg/docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion

Requirement already satisfied: fastapi in /usr/local/lib/python3.11/site-packages
(0.121.0)

Requirement already satisfied: uvicorn in /usr/local/lib/python3.11/site-packages
(0.38.0)

Requirement already satisfied: requests in /usr/local/lib/python3.11/site-
packages (2.32.5)

Requirement already satisfied: pydantic in /usr/local/lib/python3.11/site-
packages (2.12.4)

Requirement already satisfied: starlette<0.50.0,>=0.40.0 in
/usr/local/lib/python3.11/site-packages (from fastapi) (0.49.3)

Requirement already satisfied: typing-extensions>=4.8.0 in
/usr/local/lib/python3.11/site-packages (from fastapi) (4.15.0)

Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.11/site-packages (from fastapi) (0.0.3)

Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.11/site-
packages (from uvicorn) (8.3.0)

Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.11/site-
packages (from uvicorn) (0.16.0)

Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.11/site-packages (from requests) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/site-
packages (from requests) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.11/site-packages (from requests) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.11/site-packages (from requests) (2025.10.5)
```

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.11/site-packages (from pydantic) (0.7.0)

Requirement already satisfied: pydantic-core==2.41.5 in /usr/local/lib/python3.11/site-packages (from pydantic) (2.41.5)

Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.11/site-packages (from pydantic) (0.4.2)

Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.11/site-packages (from starlette<0.50.0,>=0.40.0->fastapi) (4.11.0)

Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/site-packages (from anyio<5,>=3.6.2->starlette<0.50.0,>=0.40.0->fastapi) (1.3.1)

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 24.0 -> 25.3

[notice] To update, run: pip install --upgrade pip

root@debianMSI:/home/cmateos/tfg#

Se corrige el Docker compose

```
cmateos@debianMSI ~
sudo su -
    model: str | None = "qwen2.5:7b"

@app.get("/status")
def status():
    # prueba conexión a Ollama también
    try:
        r = requests.get(f"{OLLAMA_BASE}/api/tags", timeout=5)
        ok = r.status_code == 200
    except Exception:
        ok = False
    return {"status": "ok", "ollama_ok": ok}

@app.post("/chat")
def chat(body: ChatIn):
    payload = {
        "model": body.model,
        "messages": [{"role": "user", "content": body.message}],
    }
    try:
        r = requests.post(f"{OLLAMA_BASE}/api/chat", json=payload, timeout=60)
        r.raise_for_status()
        return r.json()
    except requests.exceptions.RequestException as e:
        # Devuelve 502 si Ollama no responde o hay error de red/HTTP
        raise HTTPException(status_code=502, detail=f"Ollama unreachable: {e}")
root@debianMSI:/home/cmateos/tfg# nano docker-compose.yml
root@debianMSI:/home/cmateos/tfg# docker compose up -d --force-recreate api
WARN[0000] /home/cmateos/tfg/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it
to avoid potential confusion
[+] Running 1/1
 ✔ Container fastapi  Started                                                    0.6s
root@debianMSI:/home/cmateos/tfg# curl -s http://localhost:8000/status
curl -s -X POST http://localhost:8000/chat \
  -H "Content-Type: application/json" \
  -d '{"message":"Hola, ¿quién eres?"}' | jq
{"status":"ok","ollama_ok":true}-bash: jq: orden no encontrada
root@debianMSI:/home/cmateos/tfg#
```

Se debe instalar js ya que vemos que status ok ollama ok pero no encunetra el jq para formato JSON

```
Installing dependencies:
  libjq1  libonig5

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 0
  Download size: 431 kB
  Space needed: 1.265 kB / 899 GB available

Des:1 http://deb.debian.org/debian trixie/main amd64 libonig5 amd6
Des:2 http://deb.debian.org/debian trixie/main amd64 libjq1 amd64
Des:3 http://deb.debian.org/debian trixie/main amd64 jq amd64 1.7.
Descargados 431 kB en 0s (11,8 MB/s)
Seleccionando el paquete libonig5:amd64 previamente no seleccionad
(Leyendo la base de datos ... 69037 ficheros o directorios instala
Preparando para desempaquetar .../libonig5_6.9.9-1+b1_amd64.deb ..
Desempaquetando libonig5:amd64 (6.9.9-1+b1) ...
Seleccionando el paquete libjq1:amd64 previamente no seleccionado.
Preparando para desempaquetar .../libjq1_1.7.1-6+deb13u1_amd64.deb
Desempaquetando libjq1:amd64 (1.7.1-6+deb13u1) ...
Seleccionando el paquete jq previamente no seleccionado.
Preparando para desempaquetar .../jq_1.7.1-6+deb13u1_amd64.deb ...
Desempaquetando jq (1.7.1-6+deb13u1) ...
Configurando libonig5:amd64 (6.9.9-1+b1) ...
Configurando libjq1:amd64 (1.7.1-6+deb13u1) ...
Configurando jq (1.7.1-6+deb13u1) ...
Procesando disparadores para man-db (2.13.1-1) ...
Procesando disparadores para libc-bin (2.41-12) ...
root@debianMSI:/home/cmateos/tfg#
```

```
cmateos@debianMSI ~/tfg/api  (0.036s)
# Comprueba salud
curl -s http://localhost:8000/status

# Chat (ahora debería responder usando /api/generate si /api/chat no existe)
curl -s -X POST http://localhost:8000/chat \
  -H "Content-Type: application/json" \
  -d '{"message":"Hola, ¿quién eres?"}'

{"status":"ok","ollama_ok":true,"ollama_url":"http://ollama:11434"}{"detail":"Ollama unrea
chable: 404 Client Error: Not Found for url: http://ollama:11434/api/generate"}

cmateos@debianMSI ~/tfg/api  (0.026s)
docker exec -it ollama ollama pull qwen2.5:7b

permission denied while trying to connect to the Docker daemon socket at unix:///var/run/d
ocker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.51/containers/ollama/json": dial uni
x /var/run/docker.sock: connect: permission denied

cmateos@debianMSI ~/tfg/api
sudo docker exec -it ollama ollama pull qwen2.5:7b

pulling manifest
pulling 2bada8a74506:  48%  |  2.2 GB/4.7 GB  298 MB/s      8s
```

✅ Estado actual

Componente	Estado	Descripción
Debian 13	🟢 OK	Host estable y actualizado
Docker + red `ai_net`	🟢 OK	Infraestructura de contenedores
Ollama (GPU RTX 3060 Ti)	🟢 OK	Responde vía `/api/generate` con CUDA
FastAPI (API propia)	🟢 OK	Endpoint `/chat` y `/status` funcionales
GPU Runtime NVIDIA	🟢 OK	100 % operativo con drivers 550.x
Rendimiento	⚡ Excelente	Ryzen 5700X + 32 GB RAM + GPU → < 2 s de latencia
Logs y control	🟠 Parcial	Zabbix/Kuma aún por configurar